NUR 774 Chapter 38 Cardiovascular: 100 Hard Questions

45 Select All That Apply + 55 single answer, grouped by topic, from Chapter 38 (heart sounds, murmurs, diagnostics, primary care, specific defects, sudden cardiac death). Each question is followed by its answer, rationale, and a mnemonic.

> Heads-up: Chapter 38 says delay live vaccines 6 months after bypass/blood products. That is different from the 11-month wait after Kawasaki IVIG (2 g/kg). Know both.

How to use: answer the question, then swipe to the next page for the answer, rationale, and mnemonic.

SINGLE ANSWER

An aortic ejection click from aortic stenosis is heard BEST at which location?

A. Left infraclavicular area

B. Erb point (left sternal border, 2nd–3rd intercostal space)

C. Apex

D. Lower left sternal border

Swipe for answer →

Answer 1

B

- **Erb point (left sternal border, 2nd–3rd intercostal space)**

Why: The aortic ejection click is constant in intensity and best heard at Erb point. Erb point also picks up AS and a dilated aortic root.

> 🧠 **"Erb = Ejection click of the Aorta."**

SELECT ALL THAT APPLY

Which conditions are heard at the UPPER LEFT sternal border (ULSB)?

A. Pulmonic valve click of pulmonary stenosis

B. Mitral regurgitation

C. Patent ductus arteriosus

D. Pulmonary flow murmur

E. Ventricular septal defect

F. Atrial septal defect

Swipe for answer →

Answer 2

A, C, D, F

- **Pulmonic valve click of pulmonary stenosis**
- **Patent ductus arteriosus**
- **Pulmonary flow murmur**
- **Atrial septal defect**

Why: ULSB = pulmonic area: PS clicks, pulmonary flow murmurs, ASD, PDA, and venous hum. VSD is at the LLSB; mitral regurgitation is at the apex.

> 🧠 **ULSB = the "P-A-P" corner: Pulmonic stenosis, ASD, PDA.**

SELECT ALL THAT APPLY

Which conditions are heard best at the LOWER LEFT sternal border (LLSB)?

A. Still murmur

B. Venous hum

C. Ventricular septal defect

D. Tricuspid regurgitation

E. Aortic valve click of aortic stenosis

F. Subaortic stenosis

G. Hypertrophic cardiomyopathy

Swipe for answer →

Answer 3

A, C, D, F, G

- **Still murmur**
- **Ventricular septal defect**
- **Tricuspid regurgitation**
- **Subaortic stenosis**
- **Hypertrophic cardiomyopathy**

Why: LLSB: VSD, Still murmur, tricuspid regurgitation, HCM, and subaortic stenosis. Venous hum is in the upper chest; the AS valve click is at the URSB.

> 🧠 **LLSB = "V-STiHS": VSD, Still, Tricuspid regurg, HCM, Subaortic.**

SINGLE ANSWER

A murmur of mitral valve regurgitation is heard best at the:

A. Upper left sternal border

B. Apex

C. Upper right sternal border

D. Erb point

Swipe for answer →

Answer 4

B

- **Apex**

Why: The apex is the mitral area: mitral regurgitation and mitral/aortic clicks.

> 🧠 **"Mitral = the tip (apex) of the heart."**

SINGLE ANSWER

A 12-year-old's harsh grade III–IV/VI crescendo-decrescendo systolic murmur is loudest at the URSB and radiates to the neck. The most likely lesion is:

A. Pulmonic stenosis

B. Ventricular septal defect

C. Aortic stenosis

D. Still murmur

Swipe for answer →

Answer 5

C

- **Aortic stenosis**

Why: AS murmur: harsh, URSB, radiates to the neck. A thrill may be felt at the suprasternal notch.

> 🧠 **"AS goes up and to the right, then into the neck."**

SINGLE ANSWER

Pulmonic stenosis's systolic ejection murmur classically radiates to:

A. The neck only

B. The back and both lung fields

C. The right axilla only

D. Nowhere; it never radiates

Swipe for answer →

Answer 6

B

- **The back and both lung fields**

Why: PS: harsh mid-to-late SEM at the ULSB, radiating along the LSB, to the neck, back, and both lung fields.

> 🧠 **"PS spreads to the Pulmonary fields (lungs and back)."**

SINGLE ANSWER

Coarctation of the aorta's systolic ejection murmur is best heard in the:

A. Apex, radiating to the axilla

B. URSB, radiating to the neck

C. Left infraclavicular region, radiating to the back

D. LLSB without radiation

Swipe for answer →

Answer 7

C

- **Left infraclavicular region, radiating to the back**

Why: The CoA murmur is heard below the left clavicle, with transmission to the back (where the narrowed aorta is).

> 🧠 **"Coarct = Collarbone (left) to the back."**

SINGLE ANSWER

In a healthy child, how does S2 behave with breathing?

A. It is always single

B. It is widely split and fixed

C. It splits with inspiration and becomes single with expiration

D. It splits only with expiration

Swipe for answer →

Answer 8

C

- **It splits with inspiration and becomes single with expiration**

Why: On inspiration, more blood returns to the right side, so P2 lags behind A2 and S2 splits. On expiration it becomes single.

> 🧠 **"Breathe IN → S2 splits IN two."**

SINGLE ANSWER

Where is S2 best assessed?

A. Upper left sternal border (pulmonic area)

B. Upper right sternal border

C. Apex

D. Lower left sternal border

Swipe for answer →

Answer 9

A

- **Upper left sternal border (pulmonic area)**

Why: S2 splitting is one of the most important parts of the cardiac exam and is best heard in the pulmonic area (ULSB).

> 🧠 **"S2 → the Second rib space, left."**

SELECT ALL THAT APPLY

A child has a LOUD, single S2. Which conditions could cause this?

A. Pulmonary atresia

B. Pulmonary hypertension

C. Atrial septal defect

D. Innocent Still murmur

E. Transposition of the great arteries

Swipe for answer →

Answer 10

A, B, E

- **Pulmonary atresia**
- **Pulmonary hypertension**
- **Transposition of the great arteries**

Why: Pulmonary HTN closes P2 early and accentuates S2. A missing semilunar valve (pulmonary atresia) gives a single S2. In d-TGA, S2 is loud and single because of where the great arteries sit. ASD gives a WIDE FIXED split.

> 🧠 **"Loud and Lonely S2 = pressure (pulm HTN) or a missing/moved valve."**

SINGLE ANSWER

A wide split S2 that does NOT become single on expiration suggests:

A. Aortic stenosis

B. Pulmonary hypertension

C. A normal finding in toddlers

D. Increased pulmonary blood flow, as in an atrial septal defect

Swipe for answer →

Answer 11

D

- **Increased pulmonary blood flow, as in an atrial septal defect**

Why: Extra right-sided volume delays P2 in every breath, so the split stays wide and fixed.

> 🧠 **"ASD = Always Split, Doesn't change."**

SELECT ALL THAT APPLY

Which statements about S3 are correct?

A. It is best heard at the apex with the bell

B. It is always pathologic

C. It is easier to hear in the left lateral decubitus position

D. It is caused by rapid ventricular filling

E. It may be normal in a quiet infant with a fast heart rate

F. It sounds like "Ken-TUCK-y"

Swipe for answer →

Answer 12

A, C, D, E, F

- **It is best heard at the apex with the bell**
- **It is easier to hear in the left lateral decubitus position**
- **It is caused by rapid ventricular filling**
- **It may be normal in a quiet infant with a fast heart rate**
- **It sounds like "Ken-TUCK-y"**

Why: S3 = early diastole, rapid filling, can be normal in children. S4 is the one that is ALWAYS pathologic.

> 🧠 **"S3 = Kentucky: can be a normal Kid. S4 = Tennessee: Trouble."**

SELECT ALL THAT APPLY

Which statements about S4 are correct?

A. It is low-pitched and heard best at the apex with the bell

B. It is a normal finding in quiet infants

C. It is best heard with the diaphragm at the ULSB

D. It is always pathologic

E. It occurs in late diastole just before S1

F. It sounds like "TEN-nessee"

Swipe for answer →

Answer 13

A, D, E, F

- **It is low-pitched and heard best at the apex with the bell**
- **It is always pathologic**
- **It occurs in late diastole just before S1**
- **It sounds like "TEN-nessee"**

Why: S4 reflects forceful atrial contraction into a stiff/distended ventricle. Always pathologic.

> 🧠 **"S4 = Tennessee = Trouble, every time."**

SINGLE ANSWER

A gallop that sounds like "Kentucky" occurs during which part of the cardiac cycle?

A. Early diastole

B. Late diastole, just before S1

C. Early systole, immediately after S1

D. Midsystole

Swipe for answer →

Answer 14

A

- **Early diastole**

Why: S3 is early diastole (right after S2). S4 is late diastole (right before S1).

> 🧠 **"Ken-TUCK-y: the extra beat TUCKs in right AFTER S2."**

SELECT ALL THAT APPLY

Which statements describe a PULMONIC ejection click?

A. It occurs early in systole, right after S1

B. It is constant in intensity

C. It varies with respiration

D. It is high in frequency

E. It disappears with inspiration

F. It is heard best at the apex in midsystole

Swipe for answer →

Answer 15

A, C, D, E

- **It occurs early in systole, right after S1**
- **It varies with respiration**
- **It is high in frequency**
- **It disappears with inspiration**

Why: Pulmonic click varies with breathing and disappears on inspiration. The aortic click is the constant one.

> 🧠 **"Pulmonic click Plays with breathing. Aortic click is Always the same."**

SINGLE ANSWER

A teen has a MIDSYSTOLIC click at the apex that is loudest when standing or leaning forward. The most likely cause is:

A. Pulmonic stenosis

B. Patent ductus arteriosus

C. Mitral valve prolapse

D. Aortic stenosis

Swipe for answer →

Answer 16

C

- **Mitral valve prolapse**

Why: Nonejection clicks are midsystolic at the apex, best heard leaning forward or standing, and are due to MVP.

> 🧠 **"MVP = Middle click, apex, Very upright (standing)."**

SINGLE ANSWER

Why might a child with a LARGE septal defect have NO murmur?

A. There may be little turbulent flow across a large, nonrestrictive opening

B. Murmurs are only heard with small hearts

C. Large defects always close by 6 months

D. The stethoscope can't pick up septal defects

Swipe for answer →

Answer 17

A

- **There may be little turbulent flow across a large, nonrestrictive opening**

Why: Murmurs come from turbulence. Large nonrestrictive defects (or a nonrestrictive PDA) may not create turbulence.

> 🧠 **"No noise ≠ no problem."**

SINGLE ANSWER

A murmur is loud, heard over a wide area, with a palpable thrill, but CANNOT be heard with the stethoscope barely touching the chest. Grade?

A. V/VI

B. IV/VI

C. VI/VI

D. III/VI

Swipe for answer →

Answer 18

B

- **IV/VI**

Why: IV = loud + thrill. V = heard with the stethoscope barely on the chest. VI = heard without a stethoscope.

> 🧠 **"4 = Feel it (thrill). 5 = barely touching. 6 = no touching."**

SELECT ALL THAT APPLY

Which statements about DIASTOLIC murmurs are correct?

A. They occur between S1 and S2

B. A decrescendo murmur starting with S2 is often aortic or pulmonic regurgitation

C. Most innocent murmurs are diastolic

D. They always indicate cardiac pathology

E. A mid-diastolic rumble at the apex can come from increased flow with a large VSD or PDA

Swipe for answer →

Answer 19

B, D, E

- **A decrescendo murmur starting with S2 is often aortic or pulmonic regurgitation**
- **They always indicate cardiac pathology**
- **A mid-diastolic rumble at the apex can come from increased flow with a large VSD or PDA**

Why: Diastolic murmurs occur between S2 and S1 and are always pathologic.

> 🧠 **"Diastolic is always Dangerous."**

SINGLE ANSWER

What is the most common cause of a CONTINUOUS murmur in children?

A. Patent ductus arteriosus

B. Still murmur

C. Atrial septal defect

D. Aortic stenosis

Swipe for answer →

Answer 20

A

- **Patent ductus arteriosus**

Why: A continuous murmur runs from S1 through systole AND diastole. PDA is the most common cause; it must be told apart from a venous hum.

> 🧠 **"Continuous Machinery = PDA."**

SINGLE ANSWER

A continuous murmur below the clavicles disappears when the child lies down or turns the head. This is most likely:

A. Venous hum

B. Patent ductus arteriosus

C. Aortic regurgitation

D. Coarctation

Swipe for answer →

Answer 21

A

- **Venous hum**

Why: A venous hum is innocent and continuous, but disappears with head turning, jugular compression, or lying supine. A PDA murmur cannot be suppressed.

> 🧠 **"Hum goes Home when you lie down."**

SELECT ALL THAT APPLY

Which features are typical of an INNOCENT murmur (Box 38.2)?

A. Grade I–III and localized

B. Musical or vibratory quality

C. Associated with a palpable thrill

D. Normal vital signs and ECG

E. Short and peaking in the first half of systole

F. Pansystolic timing

G. Changes with position

Swipe for answer →

Answer 22

A, B, D, E, G

- **Grade I–III and localized**
- **Musical or vibratory quality**
- **Normal vital signs and ECG**
- **Short and peaking in the first half of systole**
- **Changes with position**

Why: Innocent murmurs are soft, short, systolic, positional, musical, and occur in an otherwise healthy child.

> 🧠 **"7 S's": Soft, Systolic, Short, Single, Small area, Sweet, Sensitive (changes with position).**

SELECT ALL THAT APPLY

Which findings suggest a POSSIBLY PATHOLOGIC murmur?

A. Grade IV/VI or louder

B. A murmur that does not change with position

C. Harsh quality

D. Musical quality that disappears with Valsalva

E. A murmur in a child with trisomy 21

F. Any diastolic murmur

Swipe for answer →

Answer 23

A, B, C, E, F

- **Grade IV/VI or louder**
- **A murmur that does not change with position**
- **Harsh quality**
- **A murmur in a child with trisomy 21**
- **Any diastolic murmur**

Why: Box 38.2 pathologic list includes diastolic, thrill, pansystolic, nonsuppressible continuous, clicks, snaps, fixed/accentuated S2, S4, non-positional, grade ≥ IV, harsh, or a syndrome linked to CHD.

> 🧠 **"Harsh, Loud, Diastolic, Doesn't move, or Down syndrome → refer."**

SINGLE ANSWER

A 4-year-old has a grade II midsystolic, vibratory, "twangy string" murmur at the LLSB that is louder lying down and softer standing. This is:

A. Pulmonary flow murmur of infancy

B. Small VSD

C. Still murmur

D. Venous hum

Swipe for answer →

Answer 24

C

- **Still murmur**

Why: Still murmur: ages 2–6, LLSB/apex, musical/vibratory, louder supine, decreases standing, does not radiate.

> 🧠 **"Still = Strings (twangy guitar), Supine louder."**

SINGLE ANSWER

A preterm newborn has a soft murmur from the left upper sternal border radiating to both axillae and the back. It is expected to:

A. Get louder throughout childhood

B. Require PDA ligation

C. Indicate tetralogy of Fallot

D. Disappear by about 6 months of age

Swipe for answer →

Answer 25

D

- **Disappear by about 6 months of age**

Why: Pulmonary flow murmur of infancy (peripheral pulmonary stenosis) is common in newborns, especially preterm, radiates to the axillae and back, and is usually gone by 6 months.

> 🧠 **"PPS = Preemie Pulmonary Swish, gone by 6 months."**

SINGLE ANSWER

A thin-chested 12-year-old has a soft, blowing systolic murmur at the pulmonic area with a normally split S2 and no click or thrill. This is most likely:

A. Atrial septal defect

B. Pulmonary flow murmur of childhood

C. Pulmonic stenosis

D. Venous hum

Swipe for answer →

Answer 26

B

- **Pulmonary flow murmur of childhood**

Why: Pulmonary flow murmur of childhood: thin-chested kids 8–14, soft blowing, normal split S2, no click or thrill. ASD would have a FIXED split; PS would have a click.

> 🧠 **"Thin teen, soft blow, Normal split = Normal flow."**

SINGLE ANSWER

A primary care provider plans to refer a child to pediatric cardiology for a murmur. What does the text recommend about ordering routine tests first?

A. Order a CBC, ECG, and MRI first

B. Order an echocardiogram first, then refer

C. Always order an ECG and chest x-ray before referral

D. Routine tests before referral are not cost-effective; cardiology decides which tests to order

Swipe for answer →

Answer 27

D

- **Routine tests before referral are not cost-effective; cardiology decides which tests to order**

Why: The cardiologist can decide more precisely which tests are needed.

> 🧠 **"Referring? Don't pre-test, let the expert pick."**

SINGLE ANSWER

A cyanotic newborn receives 100% oxygen and pinks up, with a rising saturation. This suggests:

A. The cause is primarily pulmonary

B. The test is invalid in newborns

C. The cause is primarily cardiac

D. Tetralogy of Fallot

Swipe for answer →

Answer 28

A

- **The cause is primarily pulmonary**

Why: Hyperoxia test: pinking with 100% O2 = lung disease. Minimal or no improvement = heart disease (the blood is bypassing the lungs).

> 🧠 **"Pinks up = Pulmonary. Stays blue = Heart (Cardiac)."**

SELECT ALL THAT APPLY

Which can be done during cardiac catheterization?

A. Deploy devices to close holes or extra vessels

B. Open narrowed valves or arteries

C. Diagnose CHD at 10 weeks' gestation

D. Obtain a heart tissue sample

E. Measure pulmonary vascular resistance and cardiac output

F. Screen for hearing loss

Swipe for answer →

Answer 29

A, B, D, E

- **Deploy devices to close holes or extra vessels**
- **Open narrowed valves or arteries**
- **Obtain a heart tissue sample**
- **Measure pulmonary vascular resistance and cardiac output**

Why: Cath = diagnostic AND interventional.

> 🧠 **"Cath can Measure, Biopsy, Open, and Plug."**

SINGLE ANSWER

How early can fetal echocardiography diagnose congenital heart disease?

A. Only after birth

B. 16–18 weeks' gestation (as early as 10 weeks with high-frequency transvaginal echo)

C. At 4 weeks

D. At 32 weeks

Swipe for answer →

Answer 30

B

- **16–18 weeks' gestation (as early as 10 weeks with high-frequency transvaginal echo)**

Why: Fetal echo detects CHD, arrhythmias, and hemodynamic changes before birth.

> 🧠 **"Sweet 16 for the fetal heart."**

SINGLE ANSWER

Which ECG pattern is classic for an atrial septal defect?

A. Left ventricular hypertrophy with inverted T waves

B. Superior axis between −40 and −160 degrees

C. Right axis deviation with an rSR′ (right bundle branch block) pattern in V1

D. Delta waves

Swipe for answer →

Answer 31

C

- **Right axis deviation with an rSR′ (right bundle branch block) pattern in V1**

Why: ASD → right-sided volume load: RAD, RAE, RBBB rSR′ in V1. Superior axis = AV septal defect.

> 🧠 **"ASD loads the Right: Right axis, Right bundle."**

SINGLE ANSWER

Which ECG finding is characteristic of an AV septal (endocardial cushion) defect?

A. Right bundle branch block only

B. Normal axis with LVH only

C. Prolonged QT

D. Superior axis (−40 to −160 degrees)

Swipe for answer →

Answer 32

D

- **Superior axis (−40 to −160 degrees)**

Why: AVSD shows a superior QRS axis; RVH is common and the PR is prolonged in 50%.

> 🧠 **"AV canal points UP (superior axis)."**

SELECT ALL THAT APPLY

An infant with CHF is not gaining weight. Which nutrition strategies are appropriate?

A. Diluting formula to reduce fluid volume

B. Formula or fortified breast milk at 24, 27, or 30 kcal/oz

C. Nasogastric tube feeding if the infant fatigues with feeding

D. Limiting feeds to every 6 hours

E. Referral to a nutritionist

Swipe for answer →

Answer 33

B, C, E

- **Formula or fortified breast milk at 24, 27, or 30 kcal/oz**
- **Nasogastric tube feeding if the infant fatigues with feeding**
- **Referral to a nutritionist**

Why: Infants with CHF burn more calories and tire with feeding: increase caloric density and use an NG or G tube if needed.

> 🧠 **"Pack the calories, not the volume: 24-27-30."**

SINGLE ANSWER

An infant had cardiopulmonary bypass surgery and received blood products. When can MMR and varicella be given?

A. Immediately; there is no delay

B. 11 months after surgery

C. 6 months after bypass and blood/plasma exposure

D. Never

Swipe for answer →

Answer 34

C

- **6 months after bypass and blood/plasma exposure**

Why: Ch 38: delay live vaccines 6 months after bypass and RBC/plasma exposure. Other vaccines stay on schedule. (Compare: after 2 g/kg IVIG for Kawasaki, the wait is 11 months.)

> 🧠 **"Bypass → wait 6 for live. IVIG (Kawasaki) → wait 11."**

SELECT ALL THAT APPLY

Which infants qualify for RSV prophylaxis per the chapter?

A. An infant with an innocent Still murmur

B. A healthy 3-year-old with a repaired small ASD

C. Infants under 1 year with CHD and pulmonary hypertension

D. Infants under 1 year with cyanotic CHD

E. Infants under 1 year with complicated CHD and CHF

Swipe for answer →

Answer 35

C, D, E

- **Infants under 1 year with CHD and pulmonary hypertension**
- **Infants under 1 year with cyanotic CHD**
- **Infants under 1 year with complicated CHD and CHF**

Why: AAP: RSV prophylaxis for infants younger than 1 year with cyanotic or complicated CHD, especially with CHF or pulmonary HTN.

> 🧠 **"Under 1, Blue or Burdened (CHF/pulm HTN) → RSV shield."**

SINGLE ANSWER

What is the RSV prophylaxis regimen described in the chapter?

A. 50 mg/kg PO daily

B. 15 mg/kg IM every 30 days during RSV season, maximum 5 doses

C. 15 mg/kg IV once

D. 5 mg/kg IM weekly, maximum 15 doses

Swipe for answer →

Answer 36

B

- **15 mg/kg IM every 30 days during RSV season, maximum 5 doses**

Why: Maximum 5 doses of 15 mg/kg IM monthly during the season.

> 🧠 **"15 every 30, up to 5."**

SINGLE ANSWER

A 45-year-old grandmother will babysit an infant with CHD. Her last Td was 3 years ago. What should she receive?

A. MMR only

B. Td booster in 7 years

C. Nothing; Td is still valid

D. Tdap now, regardless of the interval since her last Td

Swipe for answer →

Answer 37

D

- **Tdap now, regardless of the interval since her last Td**

Why: Anyone older than 19 spending time with infants should get Tdap, regardless of the Td interval, to prevent pertussis.

> 🧠 **"Cocoon the baby: Tdap for every adult around them."**

SELECT ALL THAT APPLY

Which statements about physical activity for children with heart disease are correct?

A. If the child can talk comfortably during activity, the intensity is likely appropriate

B. Children with ventricular arrhythmias may need complete activity restriction

C. Most children self-limit their activity

D. All children with CHD must avoid all sports

E. Cardiology should be consulted before strenuous sports

Swipe for answer →

Answer 38

A, B, C, E

- **If the child can talk comfortably during activity, the intensity is likely appropriate**
- **Children with ventricular arrhythmias may need complete activity restriction**
- **Most children self-limit their activity**
- **Cardiology should be consulted before strenuous sports**

Why: Encourage daily activity. The "talk test" guides intensity. Specific conditions need specific limits.

> 🧠 **"Talk test: if you can talk, you can walk (and play)."**

SELECT ALL THAT APPLY

Which conditions qualify for dental endocarditis prophylaxis (Box 38.3)?

A. Prosthetic heart valve

B. Unrepaired cyanotic CHD, including palliative shunts

C. CHD repaired with prosthetic material, within 6 months of repair

D. Small isolated VSD that was never repaired

E. Heart transplant recipient with valve disease

F. Previous infective endocarditis

Swipe for answer →

Answer 39

A, B, C, E, F

- **Prosthetic heart valve**
- **Unrepaired cyanotic CHD, including palliative shunts**
- **CHD repaired with prosthetic material, within 6 months of repair**
- **Heart transplant recipient with valve disease**
- **Previous infective endocarditis**

Why: Only the highest-risk conditions qualify. A small VSD does not require SBE prophylaxis.

> 🧠 **"Plastic, Past IE, Purple (cyanotic), Patch <6 months, Post-transplant valve."**

SINGLE ANSWER

A child had a residual VSD next to a surgical patch. Prophylaxis is needed:

A. Indefinitely, because there is a residual defect at or adjacent to the prosthetic patch

B. Only before tonsillectomy

C. Never

D. Only for 6 months after surgery

Swipe for answer →

Answer 40

A

- **Indefinitely, because there is a residual defect at or adjacent to the prosthetic patch**

Why: Repaired CHD with residual defects at or near a prosthetic patch or device stays high-risk.

> 🧠 **"Leftover leak by the patch = lifelong cover."**

SINGLE ANSWER

What is the oral dental prophylaxis regimen for a child without a penicillin allergy?

A. Amoxicillin 90 mg/kg/day divided BID

B. Clindamycin 20 mg/kg after the procedure

C. Amoxicillin 50 mg/kg as a single dose 30–60 minutes before the procedure

D. Amoxicillin 50 mg/kg daily for 10 days

Swipe for answer →

Answer 41

C

- **Amoxicillin 50 mg/kg as a single dose 30–60 minutes before the procedure**

Why: Single dose, 30–60 minutes BEFORE the procedure.

> 🧠 **"50 mg/kg, 1 dose, 30–60 minutes before."**

SELECT ALL THAT APPLY

A child who qualifies for prophylaxis had hives (urticaria) with amoxicillin and can take oral medications. Which choices are appropriate?

A. Azithromycin 15 mg/kg

B. Clindamycin 20 mg/kg

C. Cephalexin 50 mg/kg

D. Amoxicillin 50 mg/kg

E. Ampicillin 50 mg/kg IM

Swipe for answer →

Answer 42

A, B

- **Azithromycin 15 mg/kg**
- **Clindamycin 20 mg/kg**

Why: Cephalosporins should NOT be used with a history of anaphylaxis, angioedema, or urticaria to penicillins.

> 🧠 **"Hives to Pen? Skip the Cephs: Clinda or Azithro."**

SELECT ALL THAT APPLY

A child needing dental prophylaxis cannot take oral medication and has no penicillin allergy. Which are acceptable?

A. Azithromycin 15 mg/kg PO

B. Cefazolin or ceftriaxone 50 mg/kg IM or IV

C. Clindamycin 20 mg/kg PO

D. Ampicillin 50 mg/kg IM or IV

E. Amoxicillin 50 mg/kg PO

Swipe for answer →

Answer 43

B, D

- **Cefazolin or ceftriaxone 50 mg/kg IM or IV**
- **Ampicillin 50 mg/kg IM or IV**

Why: Non-oral options without allergy: ampicillin, cefazolin, or ceftriaxone.

> 🧠 **"No mouth? Amp or a Ceph in the muscle or vein."**

SELECT ALL THAT APPLY

Which findings should prompt referral to pediatric cardiology?

A. Oxygen saturation less than 95%

B. A pathologic murmur

C. Symptoms of CHF

D. A hard-to-classify murmur in a child with poor growth

E. A Still murmur in a thriving 4-year-old

Swipe for answer →

Answer 44

A, B, C, D

- **Oxygen saturation less than 95%**
- **A pathologic murmur**
- **Symptoms of CHF**
- **A hard-to-classify murmur in a child with poor growth**

Why: These are the chapter's findings suggestive of cardiac disease.

> 🧠 **"Under 95, Failing, Funny murmur, or Failing to grow."**

SINGLE ANSWER

An otherwise well 6-year-old has a new possibly pathologic murmur. How quickly should they be referred?

A. Only if symptoms develop

B. Within 1 hour

C. Timely but not urgent

D. Immediately to the emergency department

Swipe for answer →

Answer 45

C

- **Timely but not urgent**

Why: A child with a murmur who is otherwise doing well is referred in a timely but not urgent manner.

> 🧠 **"Well kid + murmur = soon, not siren."**

SELECT ALL THAT APPLY

Which presentations require referral AS SOON AS POSSIBLE?

A. A newborn with cyanosis and poor feeding

B. A child with an arrhythmia and dizziness

C. A well child whose innocent murmur gets louder with fever

D. A teen with chest pain during exertion

E. A child with syncope

Swipe for answer →

Answer 46

A, B, D, E

- **A newborn with cyanosis and poor feeding**
- **A child with an arrhythmia and dizziness**
- **A teen with chest pain during exertion**
- **A child with syncope**

Why: Exertional chest pain, syncope, arrhythmia, dyspnea, CHF signs, abnormal vital signs, or a symptomatic newborn → ASAP.

> 🧠 **"Pain with play, Passing out, Palpitations, Purple newborn = Prompt."**

SELECT ALL THAT APPLY

Which maternal exposures are linked to CHD?

A. Alcohol

B. ACE inhibitors

C. Ibuprofen or naproxen

D. Acetaminophen in normal doses

E. Retinoic acid

F. Lithium

Swipe for answer →

Answer 47

A, B, C, E, F

- **Alcohol**
- **ACE inhibitors**
- **Ibuprofen or naproxen**
- **Retinoic acid**
- **Lithium**

Why: Teratogens listed: lithium, retinoic acid, antiepileptics, NSAIDs, ACE inhibitors, TCAs, sulfonamides, sulfasalazine, tobacco, alcohol, cocaine, marijuana.

> 🧠 **"LIRA ACT: Lithium, Isotretinoin (retinoic acid), NSAIDs (ibuprofen), ACE inhibitors, Alcohol, Cocaine/cannabis, Tobacco."**

SINGLE ANSWER

A mother asks whether something she did caused her baby's heart defect. Which response best reflects the chapter?

A. "Most heart defects have no identifiable cause."

B. "It was caused by stress during pregnancy."

C. "It was definitely genetic from your side."

D. "It happened during labor."

Swipe for answer →

Answer 48

A

- **"Most heart defects have no identifiable cause."**

Why: Parents often wrongly blame themselves; most CHD has no identifiable cause. Clear information reduces needless guilt.

> 🧠 **"Not your fault: most have no known cause."**

SELECT ALL THAT APPLY

Which heart defects are associated with trisomy 21 (Down syndrome)?

A. VSD

B. Supravalvular aortic stenosis

C. ASD

D. AV septal defect

E. Tetralogy of Fallot

F. PDA

Swipe for answer →

Answer 49

A, C, D, E, F

- **VSD**
- **ASD**
- **AV septal defect**
- **Tetralogy of Fallot**
- **PDA**

Why: About 50% of children with trisomy 21 have CHD; complete AV canal is classic. Supravalvular AS = Williams.

> 🧠 **"Down = AV canal first, then the D's and TOF."**

SINGLE ANSWER

A girl with Turner syndrome (monosomy X) is most at risk for:

A. Bicuspid aortic valve and coarctation of the aorta

B. AV septal defect

C. Supravalvular aortic stenosis

D. Dextrocardia

Swipe for answer →

Answer 50

A

- **Bicuspid aortic valve and coarctation of the aorta**

Why: Turner: bicuspid AV, CoA (35%), pulmonic stenosis. Also check four-extremity BPs.

> 🧠 **"Turner Turns the aorta: Coarctation and Bicuspid valve."**

SELECT ALL THAT APPLY

Which heart defects are associated with 22q11.2 deletion (DiGeorge syndrome)?

A. Truncus arteriosus

B. Interrupted aortic arch

C. Mitral valve prolapse

D. Perimembranous VSD

E. Supravalvular aortic stenosis

F. Tetralogy of Fallot

Swipe for answer →

Answer 51

A, B, D, F

- **Truncus arteriosus**
- **Interrupted aortic arch**
- **Perimembranous VSD**
- **Tetralogy of Fallot**

Why: DiGeorge = conotruncal (outflow) defects.

> 🧠 **"CATCH-22: Cardiac outflow (TOF, Truncus, Interrupted arch)."**

SINGLE ANSWER

Williams syndrome (7q11.23 deletion) is classically associated with:

A. Transposition of the great arteries

B. AV septal defect

C. Coarctation and bicuspid aortic valve

D. Supravalvular aortic stenosis and pulmonic stenosis

Swipe for answer →

Answer 52

D

- **Supravalvular aortic stenosis and pulmonic stenosis**

Why: Williams = supravalvular AS and pulmonic stenosis.

> 🧠 **"Williams sings ABOVE the valve" (supravalvular).**

SELECT ALL THAT APPLY

An infant of a diabetic mother has the highest listed risk of which cardiac problems?

A. Cardiomyopathy

B. VSD

C. Mitral valve prolapse from Marfan

D. Transposition of the great arteries

E. Coarctation

F. Supravalvular aortic stenosis

Swipe for answer →

Answer 53

A, B, D, E

- **Cardiomyopathy**
- **VSD**
- **Transposition of the great arteries**
- **Coarctation**

Why: Infant of diabetic mother: TGA, VSD, CoA (3–5%); cardiomyopathy (10–20%).

> 🧠 **"Sugar Mom → Switched arteries (TGA) and a thick heart."**

SINGLE ANSWER

Which pairing of syndrome and heart defect is correct?

A. Trisomy 13: aortic root dilation

B. Marfan syndrome: tricuspid atresia

C. Williams syndrome: AV canal

D. Holt-Oram syndrome: ASD and VSD

Swipe for answer →

Answer 54

D

- **Holt-Oram syndrome: ASD and VSD**

Why: Holt-Oram (TBX5) = ASD/VSD (with upper-limb anomalies). Marfan = MVP and aortic root dilation. Trisomy 13 = VSD, PDA, dextrocardia.

> 🧠 **"Holt-Oram: Hands + Holes."**

SELECT ALL THAT APPLY

Which are signs of CHF in INFANTS (Box 38.4)?

A. Hepatomegaly

B. Diaphoresis

C. Poor feeding and poor weight gain

D. Tachypnea

E. Periorbital edema

F. Orthopnea

Swipe for answer →

Answer 55

A, B, C, D, E

- **Hepatomegaly**
- **Diaphoresis**
- **Poor feeding and poor weight gain**
- **Tachypnea**
- **Periorbital edema**

Why: Orthopnea and peripheral edema are listed for children and teens, not infants. Infants show periorbital edema.

> 🧠 **"Infant HF = Sweaty, Speedy, Swollen eyes, Stuck weight."**

SINGLE ANSWER

What is the LARGEST group of children with CHF?

A. Those with myocarditis

B. Those with rheumatic fever

C. Those with Kawasaki disease

D. Those with excessive left-to-right shunting through unrepaired defects

Swipe for answer →

Answer 56

D

- **Those with excessive left-to-right shunting through unrepaired defects**

Why: L→R shunts (VSD, PDA, AV canal) cause most pediatric CHF. The myocardium often copes, but volume overload causes symptoms.

> 🧠 **"Too much to the lungs = most HF in kids."**

SELECT ALL THAT APPLY

Which lesions typically cause CHF in the FIRST WEEK of life?

A. Hypoplastic left heart syndrome

B. Critical aortic stenosis

C. Small VSD

D. Rheumatic fever

E. Coarctation of the aorta

Swipe for answer →

Answer 57

A, B, E

- **Hypoplastic left heart syndrome**
- **Critical aortic stenosis**
- **Coarctation of the aorta**

Why: Duct-dependent left-sided obstructions present early as the ductus closes. VSD-related CHF appears later (weeks), after PVR falls.

> 🧠 **"Week 1 = Left-side blockages crash as the duct closes."**

SINGLE ANSWER

Which type of ASD is most common?

A. Unroofed coronary sinus

B. Ostium primum

C. Ostium secundum (mid-septum near the foramen ovale)

D. Sinus venosus

Swipe for answer →

Answer 58

C

- **Ostium secundum (mid-septum near the foramen ovale)**

Why: Secundum is most common. Sinus venosus is often linked with anomalous pulmonary venous return; primum is seen in Down syndrome; unroofed coronary sinus is rarest.

> 🧠 **"Secundum = Second door in the Middle (foramen ovale)."**

SINGLE ANSWER

In a child with an ASD, the systolic murmur is caused by:

A. Aortic stenosis

B. Mitral regurgitation

C. Increased flow across the pulmonary valve

D. Blood flowing through the atrial hole

Swipe for answer →

Answer 59

C

- **Increased flow across the pulmonary valve**

Why: The grade I–III crescendo-decrescendo murmur at the pulmonic area comes from extra flow across the pulmonary valve, not from the hole itself.

> 🧠 **"The hole is silent; the valve sings."**

SELECT ALL THAT APPLY

Which statements about ASD management are correct?

A. Exercise restriction is unnecessary

B. Device closure in the cath lab is possible if the child weighs more than 15 kg

C. Small defects found in infancy may close on their own

D. Lifelong SBE prophylaxis is required

E. Low-dose aspirin is given for 6 months after device closure

F. SBE prophylaxis is needed only for 6 months after closure

Swipe for answer →

Answer 60

A, B, C, E, F

- **Exercise restriction is unnecessary**
- **Device closure in the cath lab is possible if the child weighs more than 15 kg**
- **Small defects found in infancy may close on their own**
- **Low-dose aspirin is given for 6 months after device closure**
- **SBE prophylaxis is needed only for 6 months after closure**

Why: Large defects are closed after age 1, ideally before school entry.

> 🧠 **"ASD device: 15 kg, 6 months aspirin, 6 months prophylaxis."**

SINGLE ANSWER

An infant's VSD murmur becomes HIGHER-PITCHED over several months. This suggests:

A. The defect is getting larger

B. Endocarditis

C. The defect is getting smaller

D. Pulmonary hypertension is developing

Swipe for answer →

Answer 61

C

- **The defect is getting smaller**

Why: Smaller holes create higher-pitched, harsher jets.

> 🧠 **"Higher pitch = Hole shrinking."**

SELECT ALL THAT APPLY

Which findings are expected with a LARGE VSD?

A. Low-pitched holosystolic murmur at the LLSB

B. Diastolic rumble at the apex

C. Signs of CHF after the first weeks of life

D. Thrill along the left sternal border

E. S3 or S4 gallop if CHF is present

F. Grade I soft musical murmur that disappears standing

Swipe for answer →

Answer 62

A, B, C, D, E

- **Low-pitched holosystolic murmur at the LLSB**
- **Diastolic rumble at the apex**
- **Signs of CHF after the first weeks of life**
- **Thrill along the left sternal border**
- **S3 or S4 gallop if CHF is present**

Why: Large VSD = lots of flow (rumble at the apex from increased mitral flow) and CHF as PVR falls.

> 🧠 **"Big VSD: Low pitch, Rumble, Thrill, Failure."**

SELECT ALL THAT APPLY

An infant has a small VSD and no CHF symptoms. Which plan is correct?

A. Expect that many small defects close by about age 4

B. Monitor every 6 months during the first year

C. Lifelong activity restriction

D. SBE prophylaxis is not recommended

E. Immediate surgical closure

Swipe for answer →

Answer 63

A, B, D

- **Expect that many small defects close by about age 4**
- **Monitor every 6 months during the first year**
- **SBE prophylaxis is not recommended**

Why: 30–50% of VSDs are small and most close spontaneously by 4 years.

> 🧠 **"Small VSD: Watch, no pills, often seals by 4."**

SINGLE ANSWER

An infant with a pinhole VSD has a murmur, but the echocardiogram does not show the defect. The nurse understands:

A. Pinhole VSDs may cause a murmur yet not be visualized on echo

B. Cardiac catheterization is required

C. The murmur must be innocent

D. The echo must be repeated immediately

Swipe for answer →

Answer 64

A

- **Pinhole VSDs may cause a murmur yet not be visualized on echo**

Why: The chapter notes that in "pinhole" VSDs a murmur may be present while the defect isn't seen on echo.

> 🧠 **"Pinhole: heard, not seen."**

SINGLE ANSWER

A transcatheter device can be used to close a muscular VSD only if the child weighs more than:

A. 25 kg

B. 3 kg

C. 15 kg

D. 6 kg

Swipe for answer →

Answer 65

D

- **6 kg**

Why: VSD device > 6 kg. Compare: ASD device > 15 kg.

> 🧠 **"VSD 6, ASD 15."**

SELECT ALL THAT APPLY

A COMPLETE AV septal defect includes which components?

A. Pulmonary stenosis

B. Single (common) AV valve

C. Primum ASD

D. Overriding aorta

E. Inlet VSD

Swipe for answer →

Answer 66

B, C, E

- **Single (common) AV valve**
- **Primum ASD**
- **Inlet VSD**

Why: Absence of the endocardial cushion leads to a primum ASD, a common AV valve, and an inlet VSD.

> 🧠 **"Complete canal = Atrial hole + Ventricle hole + One valve."**

SINGLE ANSWER

An infant with trisomy 21 and a complete AV canal shows NO CHF signs but has a loud single S2, a precordial heave, minimal murmur, and desaturation with crying. This suggests:

A. The defect has closed

B. Mild disease needing no follow-up

C. An innocent murmur

D. Persistent high pulmonary vascular resistance (pulmonary hypertension)

Swipe for answer →

Answer 67

D

- **Persistent high pulmonary vascular resistance (pulmonary hypertension)**

Why: Some infants with trisomy 21 keep high neonatal PVR, so they don't overcirculate (no CHF) but develop pulmonary HTN. A quiet murmur here is NOT reassuring.

> 🧠 **"Down + quiet murmur + loud S2 = Pressure, not peace."**

SINGLE ANSWER

When is surgical correction of a COMPLETE AV canal defect usually done?

A. At school entry

B. Before 6 months of age

C. After age 5

D. Only if symptoms appear in adolescence

Swipe for answer →

Answer 68

B

- **Before 6 months of age**

Why: Early repair prevents irreversible pulmonary vascular disease. Partial AV canal is repaired in toddler/preschool years.

> 🧠 **"Complete canal: fix before 6 months."**

SINGLE ANSWER

In a term infant, when does the ductus arteriosus normally close?

A. Functionally at 6 months

B. Functionally in 12–72 hours; permanently in 2–3 weeks

C. Only after surgery

D. Permanently within 1 hour

Swipe for answer →

Answer 69

B

- **Functionally in 12–72 hours; permanently in 2–3 weeks**

Why: Functional closure 12–72 h, permanent closure 2–3 weeks (term).

> 🧠 **"Ductus: 3 days to shut, 3 weeks to seal."**

SINGLE ANSWER

Which infant is at HIGHEST risk for a PDA?

A. An infant of an older father

B. A premature infant weighing less than 1750 g

C. A post-term infant

D. A term male infant

Swipe for answer →

Answer 70

B

- **A premature infant weighing less than 1750 g**

Why: PDA frequency rises with prematurity (45–80% under 1750 g). Female:male ratio is 3:1.

> 🧠 **"Tiny preemie = Patent ductus."**

SINGLE ANSWER

Indomethacin is ordered for a 3-month-old TERM infant with a PDA. The nurse should question it because:

A. It must be given orally

B. Indomethacin is contraindicated and ineffective in term or older infants

C. It only works in girls

D. The dose is too low

Swipe for answer →

Answer 71

B

- **Indomethacin is contraindicated and ineffective in term or older infants**

Why: Indomethacin/ibuprofen closure works only in preterm infants.

> 🧠 **"Indomethacin is for Infants born early only."**

SELECT ALL THAT APPLY

Which statements about PDA findings and management are correct?

A. Families should expect lifelong activity limits

B. Surgical ligation does not require cardiopulmonary bypass

C. After the first weeks, a continuous machinery murmur is heard in the left infraclavicular area

D. Coils or plugs can close many PDAs in the cath lab in children older than 8 months

E. SBE prophylaxis is recommended for 6 months after device closure

F. The murmur is continuous immediately after birth

Swipe for answer →

Answer 72

B, C, D, E

- **Surgical ligation does not require cardiopulmonary bypass**
- **After the first weeks, a continuous machinery murmur is heard in the left infraclavicular area**
- **Coils or plugs can close many PDAs in the cath lab in children older than 8 months**
- **SBE prophylaxis is recommended for 6 months after device closure**

Why: Right after birth, the murmur is soft and systolic. It becomes continuous after the first weeks.

> 🧠 **"PDA: early whisper (systolic), later Machinery."**

SINGLE ANSWER

At approximately what oxygen saturation does cyanosis become VISIBLE?

A. 95%

B. 85%

C. 65%

D. 75%

Swipe for answer →

Answer 73

B

- **85%**

Why: Visible cyanosis appears at about 85%. It is easier to see with polycythemia and harder with anemia or fetal hemoglobin.

> 🧠 **"85 = you start to see blue."**

SELECT ALL THAT APPLY

Which statements about Tetralogy of Fallot are correct?

A. Cyanosis usually increases over the first months as RVOT obstruction worsens

B. "Pink tets" may have little cyanosis

C. It is the most common acyanotic lesion

D. It is more common with 22q11.2 deletion and Down syndrome

E. It always presents with severe cyanosis at birth

F. It is the most common cyanotic heart lesion

Swipe for answer →

Answer 74

A, B, D, F

- **Cyanosis usually increases over the first months as RVOT obstruction worsens**
- **"Pink tets" may have little cyanosis**
- **It is more common with 22q11.2 deletion and Down syndrome**
- **It is the most common cyanotic heart lesion**

Why: TOF has a spectrum of severity. Mild cases act like a large VSD, and cyanosis is usually present by 6 months.

> 🧠 **"TOF: Tops the cyanotic list, Turns bluer over time."**

SELECT ALL THAT APPLY

Which interventions are appropriate for a hypercyanotic spell?

A. Soothing the infant

B. Oxygen

C. Giving a large feeding

D. Knee-chest position

E. Subcutaneous morphine as ordered

F. Laying the infant flat and supine

Swipe for answer →

Answer 75

A, B, D, E

- **Soothing the infant**
- **Oxygen**
- **Knee-chest position**
- **Subcutaneous morphine as ordered**

Why: Knee-chest raises systemic resistance, reducing right-to-left shunting and increasing pulmonary blood flow.

> 🧠 **"Tuck, Calm, O2, Morphine (TCOM)."**

SELECT ALL THAT APPLY

Which test findings are expected in TOF?

A. Boot-shaped heart with decreased pulmonary vascular markings

B. Egg-shaped heart

C. Rising hemoglobin and hematocrit over time

D. Right ventricular hypertrophy with right axis deviation on ECG

E. Rib notching

Swipe for answer →

Answer 76

A, C, D

- **Boot-shaped heart with decreased pulmonary vascular markings**
- **Rising hemoglobin and hematocrit over time**
- **Right ventricular hypertrophy with right axis deviation on ECG**

Why: Egg on a string = TGA. Rib notching = coarctation.

> 🧠 **"Boot = TOF, Egg = TGA, Notches = Coarct."**

SELECT ALL THAT APPLY

After TOF repair, which long-term issues need lifelong cardiology follow-up?

A. Rheumatic fever

B. A widening QRS duration, which raises the risk of VT and sudden death

C. Atrial and ventricular arrhythmias

D. Coronary artery aneurysm

E. Pulmonary valve regurgitation

Swipe for answer →

Answer 77

B, C, E

- **A widening QRS duration, which raises the risk of VT and sudden death**
- **Atrial and ventricular arrhythmias**
- **Pulmonary valve regurgitation**

Why: Progressive RV dilation widens the QRS; a QRS ≥ 180 ms raises the risk of VT and sudden death. Cardiology must clear for sports.

> 🧠 **"Fixed TOF still needs: Valve, Rhythm, QRS checks."**

SINGLE ANSWER

A child with polycythemia from cyanotic heart disease is at increased risk for:

A. Cerebral thrombosis (stroke)

B. Bleeding into the joints

C. Hypoglycemia

D. Anemia

Swipe for answer →

Answer 78

A

- **Cerebral thrombosis (stroke)**

Why: Polycythemia boosts oxygen-carrying capacity but thickens the blood. Prevent dehydration.

> 🧠 **"Thick blood → brain clot."**

SELECT ALL THAT APPLY

Which findings are typical of d-transposition of the great arteries?

A. Bounding femoral pulses with wide pulse pressure

B. Egg-shaped heart on x-ray

C. Often large for gestational age

D. Loud, single S2

E. Cyanosis within the first hour to first day of life

F. Boot-shaped heart

Swipe for answer →

Answer 79

B, C, D, E

- **Egg-shaped heart on x-ray**
- **Often large for gestational age**
- **Loud, single S2**
- **Cyanosis within the first hour to first day of life**

Why: d-TGA: aorta from the RV, PA from the LV; parallel circulations. Male:female 3:1.

> 🧠 **"TGA = Turns blue Gigantic baby, Egg heart."**

SELECT ALL THAT APPLY

Which interventions are expected for a newborn with d-TGA?

A. Balloon atrial septostomy

B. Lifelong SBE prophylaxis

C. Indomethacin to close the ductus

D. Arterial switch (Jatene) procedure in the first days of life

E. IV prostaglandin E1

Swipe for answer →

Answer 80

A, B, D, E

- **Balloon atrial septostomy**
- **Lifelong SBE prophylaxis**
- **Arterial switch (Jatene) procedure in the first days of life**
- **IV prostaglandin E1**

Why: Keep mixing (PGE1, septostomy), then switch the arteries. Annual echo follow-up.

> 🧠 **"Open, Mix, Switch."**

SINGLE ANSWER

Without treatment, what is the mortality of d-TGA?

A. 100% within the first hour

B. Nearly 0% with oxygen alone

C. 10% by the first year

D. 50% in the first month and 90% by the first year

Swipe for answer →

Answer 81

D

- **50% in the first month and 90% by the first year**

Why: Untreated d-TGA is usually fatal. With surgery, long-term results are excellent.

> 🧠 **"TGA untreated: half by 1 month, nearly all by 1 year."**

SINGLE ANSWER

A child with hypoplastic left heart syndrome is undergoing staged palliation. Match the procedure to its timing:

A. Arterial switch at 4–6 months

B. Both stages within the first week

C. Fontan at birth; Glenn at 10 years

D. Bidirectional SVC-to-PA anastomosis at 4–6 months; Fontan (IVC-to-PA) at 2–4 years

Swipe for answer →

Answer 82

D

- **Bidirectional SVC-to-PA anastomosis at 4–6 months; Fontan (IVC-to-PA) at 2–4 years**

Why: Stage 1 (newborn shunt) → Glenn (SVC→PA) at 4–6 months → Fontan (IVC→PA) at 2–4 years. Transplant if staging fails.

> 🧠 **"Superior first, Inferior last: S before I, like the alphabet."**

SELECT ALL THAT APPLY

Which are long-term complications of single-ventricle palliation?

A. Decreased exercise tolerance

B. Arrhythmias

C. Rheumatic heart disease

D. Thromboembolic events, including stroke

E. Protein-losing enteropathy

Swipe for answer →

Answer 83

A, B, D, E

- **Decreased exercise tolerance**
- **Arrhythmias**
- **Thromboembolic events, including stroke**
- **Protein-losing enteropathy**

Why: Fontan physiology strains many systems. Families need lifelong support and developmental screening.

> 🧠 **"Fontan Four: Protein loss, Palpitations, Plugs (clots), Poor stamina."**

SELECT ALL THAT APPLY

In tricuspid atresia, which statements are correct?

A. The right ventricle is small

B. The tricuspid valve is widely open

C. Blood must cross an ASD to the left atrium

D. It is acyanotic

E. About half of patients also have transposition of the great arteries

Swipe for answer →

Answer 84

A, C, E

- **The right ventricle is small**
- **Blood must cross an ASD to the left atrium**
- **About half of patients also have transposition of the great arteries**

Why: No RA→RV path. Blood goes across the atrial septum, mixes, and the child is cyanotic.

> 🧠 **"Locked tricuspid → Side door (ASD) → Mixed blue blood."**

SINGLE ANSWER

When is SBE prophylaxis recommended for a child with a single-ventricle defect?

A. Never

B. While the child remains cyanotic

C. Only for 6 months after the Fontan

D. Only before tonsillectomy

Swipe for answer →

Answer 85

B

- **While the child remains cyanotic**

Why: Unrepaired or palliated cyanotic CHD (including shunts) qualifies for prophylaxis.

> 🧠 **"Blue = Bacterial cover."**

SELECT ALL THAT APPLY

Which statements about aortic stenosis are correct?

A. Valvular stenosis is the most common type

B. It causes left ventricular hypertrophy

C. It mainly causes right ventricular hypertrophy

D. It can cause ischemia and fatal ventricular arrhythmias

E. The stenotic valve is usually bicuspid

F. It is about 4 times more common in males

Swipe for answer →

Answer 86

A, B, D, E, F

- **Valvular stenosis is the most common type**
- **It causes left ventricular hypertrophy**
- **It can cause ischemia and fatal ventricular arrhythmias**
- **The stenotic valve is usually bicuspid**
- **It is about 4 times more common in males**

Why: Increased LV pressure load → LVH → oxygen demand outstrips coronary supply → ischemia and arrhythmias.

> 🧠 **"AS: Bicuspid, Boys, Big Left ventricle."**

SELECT ALL THAT APPLY

Which physical findings suggest aortic stenosis?

A. Wide pulse pressure with bounding pulses

B. Narrow pulse pressure

C. Thrill at the suprasternal notch

D. Faint early systolic click

E. Harsh grade III–IV/VI crescendo-decrescendo systolic murmur at the URSB

Swipe for answer →

Answer 87

B, C, D, E

- **Narrow pulse pressure**
- **Thrill at the suprasternal notch**
- **Faint early systolic click**
- **Harsh grade III–IV/VI crescendo-decrescendo systolic murmur at the URSB**

Why: Wide pulse pressure with bounding pulses = PDA (or aortic insufficiency). AS narrows the pulse pressure.

> 🧠 **"AS = Narrow pulse, Neck radiation, Notch thrill."**

SINGLE ANSWER

A child with aortic INSUFFICIENCY (regurgitation) would have which murmur?

A. Midsystolic click at the apex

B. Holosystolic murmur at the ULSB

C. Early diastolic blowing murmur from the LLSB to the apex

D. Continuous machinery murmur

Swipe for answer →

Answer 88

C

- **Early diastolic blowing murmur from the LLSB to the apex**

Why: Aortic regurgitation = early diastolic, decrescendo, blowing. Remember: diastolic = always pathologic.

> 🧠 **"Leaky aortic valve = Diastolic blow."**

SELECT ALL THAT APPLY

Which sports guidance for aortic stenosis is correct?

A. Severe AS: all sports are fine if there are no symptoms

B. Moderate AS: low-intensity sports such as golf or bowling, per cardiology

C. Mild AS: no sports allowed

D. Severe AS: avoid competitive or intensive sports

E. Mild AS: all sports, with annual cardiac exams

Swipe for answer →

Answer 89

B, D, E

- **Moderate AS: low-intensity sports such as golf or bowling, per cardiology**
- **Severe AS: avoid competitive or intensive sports**
- **Mild AS: all sports, with annual cardiac exams**

Why: Severe AS (or moderate with symptoms) risks sudden death from ventricular arrhythmias.

> 🧠 **"Mild = all, Moderate = mellow (golf, bowling), Severe = sit out."**

SINGLE ANSWER

A teen had mechanical aortic valve replacement. Which teaching is essential?

A. Aspirin for 6 months only

B. Lifelong anticoagulation with warfarin

C. No medications are needed

D. Avoid all vaccines

Swipe for answer →

Answer 90

B

- **Lifelong anticoagulation with warfarin**

Why: Mechanical valves are prothrombotic and require warfarin. Tissue valves have limited durability in children.

> 🧠 **"Metal valve = Warfarin forever."**

SELECT ALL THAT APPLY

Which statements about pulmonic stenosis are correct?

A. A systolic ejection click may decrease with inspiration

B. Critical PS in a newborn can cause cyanosis via right-to-left flow across the foramen ovale

C. Short stature is common when PS occurs with Noonan or Turner syndrome

D. The click gets louder with inspiration

E. Most children are asymptomatic with a murmur found on routine exam

Swipe for answer →

Answer 91

A, B, C, E

- **A systolic ejection click may decrease with inspiration**
- **Critical PS in a newborn can cause cyanosis via right-to-left flow across the foramen ovale**
- **Short stature is common when PS occurs with Noonan or Turner syndrome**
- **Most children are asymptomatic with a murmur found on routine exam**

Why: The pulmonic click decreases with inspiration and increases with expiration.

> 🧠 **"Pulmonic click quiets on the inhale."**

SINGLE ANSWER

Balloon valvuloplasty for pulmonic stenosis is indicated when the gradient is greater than:

A. 50 mm Hg

B. 100 mm Hg

C. 20 mm Hg

D. 10 mm Hg

Swipe for answer →

Answer 92

A

- **50 mm Hg**

Why: Balloon valvuloplasty for gradients > 50 mm Hg; surgery if unsuccessful.

> 🧠 **"PS: over 50 → Balloon."**

SINGLE ANSWER

A child with MILD pulmonic stenosis wants to play sports. Which guidance is correct?

A. Treat the child normally without activity limits, but monitor for progression during growth spurts

B. Repair immediately

C. Restrict all physical activity

D. Give lifelong SBE prophylaxis

Swipe for answer →

Answer 93

A

- **Treat the child normally without activity limits, but monitor for progression during growth spurts**

Why: Moderate PS can progress during rapid growth (infancy, adolescence). SBE prophylaxis is not needed unless post-op (6 months) or there is prosthetic material.

> 🧠 **"Mild PS: Play on, but recheck in growth spurts."**

SELECT ALL THAT APPLY

Which findings are consistent with coarctation of the aorta?

A. Weak or delayed femoral pulses

B. Bounding brachial, radial, and carotid pulses

C. Upper-extremity hypertension

D. Rib notching on chest x-ray

E. Leg pain or headaches with exercise

F. Cyanosis of the lips at rest

Swipe for answer →

Answer 94

A, B, C, D, E

- **Weak or delayed femoral pulses**
- **Bounding brachial, radial, and carotid pulses**
- **Upper-extremity hypertension**
- **Rib notching on chest x-ray**
- **Leg pain or headaches with exercise**

Why: Hypertension above the narrowing, hypotension below it. Collateral vessels notch the ribs.

> 🧠 **"Bounding up, Barely down, Bumpy ribs."**

SELECT ALL THAT APPLY

A 6-week-old with severe coarctation would most likely show:

A. Cool lower extremities

B. Tachypnea

C. Lower-body mottling or pallor

D. Hypertension in the legs

E. Bounding femoral pulses

F. Poor feeding

Swipe for answer →

Answer 95

A, B, C, F

- **Cool lower extremities**
- **Tachypnea**
- **Lower-body mottling or pallor**
- **Poor feeding**

Why: Severe infant CoA shows up in the first 6 weeks with heart failure and poor lower-body perfusion.

> 🧠 **"Cold legs + poor feeding baby → check femorals."**

SELECT ALL THAT APPLY

Which statements about coarctation management are correct?

A. Lifelong SBE prophylaxis is always required

B. Blood pressure must be monitored lifelong for recoarctation

C. PGE1 maintains or reopens the ductus in critical neonatal coarctation

D. Restenosis is more likely if repair occurs before 1 year of age

E. Most children with CoA have a bicuspid aortic valve

Swipe for answer →

Answer 96

B, C, D, E

- **Blood pressure must be monitored lifelong for recoarctation**
- **PGE1 maintains or reopens the ductus in critical neonatal coarctation**
- **Restenosis is more likely if repair occurs before 1 year of age**
- **Most children with CoA have a bicuspid aortic valve**

Why: SBE prophylaxis is needed only for 6 months post-op or if prosthetic material is used.

> 🧠 **"Coarct: Fix it, then Follow BP forever."**

SINGLE ANSWER

After coarctation repair, a teen may join competitive sports if:

A. The arm-leg BP gradient is less than 20 mm Hg and peak systolic BP is normal at rest and with exercise

B. The child has no headaches

C. At least 1 month has passed

D. The arm-leg gradient is less than 50 mm Hg

Swipe for answer →

Answer 97

A

- **The arm-leg BP gradient is less than 20 mm Hg and peak systolic BP is normal at rest and with exercise**

Why: Also avoid high-intensity static exercise (weight lifting, wrestling) for the first year after surgery.

> 🧠 **"Under 20 = Good to go; no heavy lifting year 1."**

SINGLE ANSWER

What is the MOST frequent cause of exercise-related sudden cardiac death in the young?

A. Kawasaki disease

B. Atrial septal defect

C. Innocent murmur

D. Hypertrophic cardiomyopathy

Swipe for answer →

Answer 98

D

- **Hypertrophic cardiomyopathy**

Why: Cardiomyopathies, especially HCM and to a lesser extent arrhythmogenic right ventricular cardiomyopathy (ARVC), are the most common causes.

> 🧠 **"HCM = Heart Collapses Midgame."**

SINGLE ANSWER

A 13-year-old catcher collapses after a baseball hits his chest. The mechanism is:

A. Aortic dissection from Marfan syndrome

B. Hypertrophic cardiomyopathy

C. Commotio cordis: a blow during the T wave triggers ventricular fibrillation

D. Rheumatic heart disease

Swipe for answer →

Answer 99

C

- **Commotio cordis: a blow during the T wave triggers ventricular fibrillation**

Why: Blunt, nonpenetrating trauma to the midchest at the wrong moment (T wave) → VF. Immediate CPR and AED.

> 🧠 **"Commotio = Chest hit at the T wave."**

SELECT ALL THAT APPLY

Which history findings suggest HCM or ARVC in an athlete?

A. Syncope

B. Family history of cardiomyopathy

C. Palpitations

D. A soft murmur that disappears when standing

E. Chest pain or dyspnea with exertion

Swipe for answer →

Answer 100

A, B, C, E

- **Syncope**
- **Family history of cardiomyopathy**
- **Palpitations**
- **Chest pain or dyspnea with exertion**

Why: HCM may have a dynamic late systolic ejection murmur (louder with standing/Valsalva) and a forceful apical beat. ARVC is more common in males around age 15.

> 🧠 **"Faint, Family, Flutters, Fatigue on effort = screen the heart."**